eering work. The characteristics of classical science, relatively stable theoretical frameworks and generally accepted standards for the production of knowledge serving as the canonical reference for a scientific community, were, in any case, not yet shared by the scientific endeavors of the Scientific Revolution.

The conceptual framework of classical science comprising basic concepts such as inertia, a new methodical canon including the experimental method and mathematical techniques such as the differential calculus, new images of knowledge such as the mechanization of the world view, and the new institutions of classical science such as the academies were nevertheless consequences of the Scientific Revolution, as a belated result of reflection on its accumulated experience.

See also: Evolution: Diffusion of Innovations; History of Science; History of Technology; Innovation, Theory of; Kuhn, Thomas S (1922–96); Physical Sciences: History and Sociology; Technological Innovation

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Scientometrics

1. Introduction

Scientometrics can be defined as the study of the quantitative aspects of scientific communication, R&D practices, and science and technology (S&T) policies. The objective is to develop indicators of the intellectual and social organization or the sciences using network relations between scientific authors and texts. The specialty has developed in relation to the increased capacities of computer storage and information retrieval of scientific communications (e.g., citation analysis). Archival records of scientific communications contain institutional address information, substantive messages (e.g., title words), and relational information from which one is able to reconstruct patterns and identify the latent characteristics of both authors and document sets. Using scientometric techniques, one is thus able to relate institutional characteristics at the level of research groups with developments at the level of scientific disciplines and specialties.

Citations, for example, can be used for retrieving documents on the basis of author names, or vice versa. The scientometric representation is formal: it remains in need of an interpretation. The focus on uncertainty contained in the distribution relates scientometrics additionally to the (neo-evolutionary) study of complex and adaptive systems. Simulation models are increasingly used for the study of the role of science-based technologies in innovation processes. However, the specialty remains data driven because of its mission to provide indicators to S&T policy processes and R&D management.

2. A Metric of Science?

In 1978, the journal *Scientometrics* was launched as a new medium to stimulate the quantitative study of scientific communication. Derek de Solla Price, one of the founding fathers of the specialty, proclaimed in his introduction the development of scientometrics as the

emergence of a 'relatively hard' social science. This claim has generated discussion from the very beginning of the specialty. In that same year (1978), leading sociologists of science published an edited volume entitled *Toward a Metric of Science: The Advent of Science Indicators*, dedicated to 'Paul F. Lazarsfeld (1901–76), Master of Quantitative and Qualitative Social Research' (Elkana et al. 1978).

The systematic comparison of science indicators across fields of science was made possible by the creation of the *Science Citation Index* by Eugene Garfield of the Institute of Scientific Information (Garfield 1979). A preliminary version of this index became available in 1962. The creation of the database has stimulated the development of new perspectives on studies in various traditions. For example, the growth of scientific disciplines and specialties can be discussed in quantitative terms using this database (e.g., Price 1963), 'invisible colleges' can be explained in terms of network structures (Crane 1969), and theories of citations can perhaps be developed and tested (cf. Braun 1998).

The development of the specialty went hand in hand with the need for a means of legitimating science policies (Wouters 1999). Narin (1976) elaborated an instrumentarium for the systematic development of the biennial series of *Science Indicators* which the National Science Foundation of the USA began providing in 1972. (In 1987, the name of this series was changed to *Science and Engineering Indicators*.) With the further development of S&T policies in other nation-states and with the gradual emergence of such policies at the European level, scientometrics became a booming business during the 1980s. By comparing radio-astronomy facilities at the international level, Martin and Irvine (1983) showed the feasibility of comparing research groups in terms of quantitative performance indicators.

In a special issue of *Social Studies of Science* about performance indicators, Collins (1985) raised the question of the unit of analysis: what is being assessed in terms of what? The authors, the papers, or the cognitions shaped in terms of sociocognitive interactions among authors and discourses? In relation to French traditions of linguistic analysis, Callon et al. (1983) proposed using words and their co-occurrences instead of citations and co-citations (Small 1973) as units of analysis. Citations can be considered as subtextual codifications, while words indicate the variation at the level of texts. Words may change both in meaning and in terms of their observable frequency distributions.

3. Methodologies

The availability of well-organized relational databases on an annual basis challenges the scientometrician to develop comparative statistics. How is the structure in each of these years to be depicted, and how may changes in structure be distinguished from and related to changes in the observable variation? Can the difference over time be identified as 'growth of science,' or is it mainly a difference between two measurement errors? How significant are differences when tested against simulation results?

In principle, the idea of a dynamic mapping of science requires an independent operationalization of structural (that is, latent) dimensions of the maps and observable variation which is to be pencilled into these maps. Science, however, develops not only in terms of the variation, but also by changing its structural dimensions. Because of the prevailing reflexivity within the science system, previous structures can be felt as constraints and at the same time be used as resources. The construction of structure may historically be stabilized, but reflexive actors are able to deconstruct and to assess the previous constructions with hindsight.

The methodological apparatus for the mapping of science in terms of multivariate statistics (multidimensional scaling, cluster analysis, etc.) has been a product of the 1980s (e.g., Van Raan 1988). The 1990s have provided the evolutionary turn: what does history mean in relation to (envisaged?) future options? How can the system itself be informed reflexively with respect to its self-organizing capacities (Leydesdorff 1995)? The relation to technometrics and the measurement of 'systems of innovation' has become central to a shifting research agenda. The development of the sciences has increasingly been contextualized in relation to science-based technologies and innovation systems (Gibbons et al. 1994).

3.1 Comparative Statistics of Science

Various methods for the mapping of science can be based on relational indicators such as citations, words, co-occurrences of each of these categories, etc. Clustering, however, requires the choice of a similarity criterion and a clustering algorithm. The distinction between positional (or factor analytical) and relational (or graph analytical) approaches is also relevant: the network is expected to contain an architecture in which actors have a position. A nearby position does not necessarily imply a relation. The methodological reflection may thus help to clarify the theoretical analysis (Burt 1982).

The mapping requires a specific perspective for the projection. Each perspective assumes a position or a window. If the multidimensional space is highly structured, the different positions may provide nearly incommensurable projections. In an article on the development of the relation between scientometrics and other subfields of S&T studies, Leydesdorff and Van den Besselaar (1997) showed that the mappings depicting science studies from the perspective of the journals *Scientometrics, Social Studies of Science*, and *Research Policy*, respectively, are increasingly different

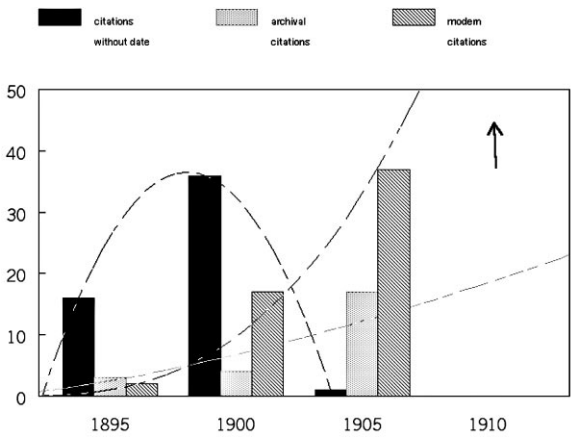


Figure 1
The emergence of the modern citation in the February issues of *Journal of the American Chemical Society* (1890–1910) (after Leydesdorff and Wouters 1999, p. 175).

during the 1990s. Citation relations among these core journals tend to decrease. The authors characterize *Social Studies of Science* as the 'codifier' of the field (along the historical axis), *Scientometrics* as the 'formalizer,' and *Research Policy* as the 'utilizer.' Only a few scholars in 'Science, Technology, and Innovation Studies' have developed competences for communicating across these subdisciplinary boundaries.

3.2 Time-series Methodologies

During the 1980s a debate raged in the community concerning the scientometric indication of a 'decline of British science.' Eventually, a special issue of *Scientometrics* was devoted to the debate in 1991 (Martin 1991). Some agreement could be reached that the inclusion and exclusion of data types *and* the framework for the comparison can be crucial for the dynamic evaluation. Should one compare with reference to a previous stage (for example, in terms of an *ex ante* fixed journal set), or should one with hindsight reconstruct the relevant data? For example, not only has the number of biotechnology journals changed, but also our understanding of 'biotechnology' is changing continuously. The sociological understanding of scientific knowledge production and control seems to have eroded Price's dream of developing scientometrics as a relatively 'hard social science.'

Using time-series analysis, one is always able to increase the fit of the curve by allowing for higher-order polynomials. Here again, the theoretical appreciation has to guide the choices of the parameters. For example, if one wishes to measure growth, it may be useful to include a second- or third-order polynomial in addition to the linear fit. Figure 1, for example, depicts the emergence of the modern citation itself as a historical phenomenon, but using theoretically informed helplines. Scientific citation emerged around the turn of the twentieth century as a means to search in both the textual and the social dimensions of science. Thus, the citation can be considered as an indicator of the complexity of sociocognitive interaction in science after its institutionalization in the nineteenth century.

3.3 Neo-evolutionary Methodologies

The networks of texts and the networks of authors operate upon each other in a selective mode. The distributions are expected to contain information, and this information may be increasingly codified by recurrent selections. As the systems 'lock-in' (in terms of their mutual information), the closure of the communication into a paradigm is one among various possibilities. The uncertainties which prevail in these networks can interactively generate codifications, which can be expected to perform a 'life-cycle.' Both participants and observers are able to hypothesize these structures reflexively, and new information can be expected to induce an update. Thus, the codification drives the knowledge production process. Codification also provides instruments for local control of otherwise global developments.

The relational operation is recursive. For example, citations refer to other texts and/or to citations in other texts. The networks resulting from this operation are expected to have an architecture (which can be mapped at each moment in time). Operations are expected to be reproduced if they are able to further the production of new knowledge and the latter's retention into cognitive structures. What is functional, however, is decided at a next moment in time, that is, at the level of a reflexive hypernetwork that overlays the historically generated ones. The distributions indicate the patterns of the expected operations. Thus, the process of new knowledge claims is propelled and made more precise and selective in tradeoffs of references in social and cognitive dimensions.

Scientometric studies can be helpful in revealing the patterns of intellectual and social organization which may have remained (partially) latent to the knowledgable actors involved. Simulation studies using scientometric mappings as input enable us to indicate the difference that the moves of the players can make. The complexity of the scientists' worlds is reflected in the scientometric reconstructions. The recognition of these objectified reconstructions recursively assumes and potentially refines the cognition within the discourses at both levels.

Over time, the cognitive reconstruction becomes thoroughly selective: citations may be 'obliterated by incorporation' into the body of knowledge, and social factors may play a role in further selections, e.g., in

terms of reputations. In this co-evolution between communications and authors, distributions of citations function, among other things, as contested boundaries between specialties. Since the indicators are distributed, the boundaries remain to be validated. Functions are expected to change when the research front moves further. By using references, authors position their knowledge claims within one specialty area or another. Some selections are chosen for stabilization, for example, when codification into citation classics occurs. Some stabilizations are selected for globalization at a next-order level, for example, when the knowledge component is integrated into a technology.

4. Conclusion

The focus on evolutionary dynamics relates scientometrics increasingly with the further development of evolutionary economics (Leydesdorff and Van den Besselaar 1994). How can systems of innovation be delineated? How can the complex dynamics of such systems be understood? How is the (potentially random) variation guided by previously codified expectations? How can explorative variation be increased in otherwise 'locked-in' trajectories of technological regimes or paradigms?

From this perspective, the indication of newness may become more important than the indication of codification. The Internet, of course, offers a research tool for what has also now been called 'sitations' (Rousseau 1997). 'Webometrics' may develop as a further extension of scientometrics relating this field with other subspecialities of science and technology studies, such as the public understanding of science or the appropriation of technology and innovation using patent statistics.

See also: Communication: Electronic Networks and Publications; History of Science; Libraries; Science and Technology, Social Study of: Computers and Information Technology; Science and Technology Studies: Experts and Expertise

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Screening and Selection

1. Introduction

In order to assist in selecting individuals possessing either desirable traits such as an aptitude for higher education or skills needed for a job or undesirable ones such as having an infection, illness or a propensity to lie, screening tests are used as a first step. A more rigorous selection device, e.g., diagnostic test or detailed interview is then used for the final classification.